

The Expression of Victory and Loss: Estimating Who's Leading or Trailing from Nonverbal Cues in Sports

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Abstract The goal of the paper was to test if humans can detect whether athletes are trailing or leading in sports based on the perception of thin slices of athletes' nonverbal behavior. In Experiment 1, participants who were unexperienced in the respective sports watched short videos depicting basketball and table tennis players and rated whether athletes were trailing or leading. Results indicated that participants could significantly differentiate between trailing and leading athletes in both team and individual sports. Experiment 2 showed that children were also able to distinguish between trailing and leading athletes based on nonverbal behavior. Comparison with the adult results from Experiment 1 revealed that the adult ratings corresponded to a higher degree with the actual scores during the game compared to the children's. In Experiment 3, we replicated the findings from Experiment 1 with both expert and unexperienced participants and a different set of stimuli from team handball. Both experts and unexpert participants were able to differentiate between leading and trailing athletes. Our findings are in line with evolutionary accounts of nonverbal behavior and suggest that humans display nonverbal signals as a consequence of leading or trailing which are reliably interpreted by others. By comparing this effect as a function of different age groups we provide evidence that although even young children can differentiate between leading and trailing athletes, the decoding of subtle nonverbal cues continues to develop with increasing experience and maturation processes.

Keywords Sport · Emotion expression · Nonverbal behavior · Thin slices · Development

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Introduction

When viewing televised coverage of sport events, it is common to hear commentators referring to the body language of competing players whilst also speculating about the mental states of the performers. For example, one often hears statements such as the following: “You can tell that the team has given up;” “from looking at their body language, one suspects they’ve lost belief in their ability to turn the game around;” or “the player is absolutely dominating her opponent.” According to evolutionary accounts of nonverbal expressions, this is not surprising as humans have evolved to be especially well-equipped for communicating important internal states nonverbally—displaying emotional nonverbal expressions and interpreting these (Darwin 1872; Ekman 1992; Shariff and Tracy 2011 for a recent review) or communicating social intentions (Fridlund 1994). For example, it has been suggested that primates send submissive nonverbal cues when losing a fight in order to avoid further potential life-threatening attacks (de Waal 1998). On the other hand, winning a fight leads primates to show dominant and proud nonverbal behaviors (NVB), which are also believed to be adaptive, as they communicate their superiority and higher status over the opponent and thereby can also save valuable resources by preventing further aggressive encounters (Archer 2006; de Waal 1998; Mazur and Booth 1998; Mehta and Josephs 2010; Mehta et al. 2008). The present research addresses the question whether leading and trailing in sport events are associated with nonverbal cues that are readily understood and interpreted as signs of leading or trailing in the game due to our evolutionary inheritance.

Functions of Nonverbal Expressions of Emotions

Charles Darwin (1872) suggested that the expression of emotion in general serves two classes of functions: (1) to prepare the organism to respond adaptively to recurring stimuli; and, of especial relevance to the present study, (2) to communicate important social information. More recent theorizing has built on Darwin’s proposal and has been termed the *two stage model of emotional expression* (see Shariff and Tracy 2011, for a review), suggesting that internal physiological regulation was probably the original adaptive function of emotional expressions and later evolved to serve communication functions. Today, the communicative function of nonverbal emotional expressions is believed to be more important in humans than the physiological function due to the increased importance of quickly and nonverbally communicating social information in complex societies (Shariff and Tracy 2011, for a review). Therefore, it has been suggested that humans have evolved both the ability to automatically display NVB and to automatically interpret and adequately respond to these NVB (Ekman 1992).

An alternative to this emotion-expression view was proposed by Fridlund (1994) who suggested that nonverbal displays are not necessarily about expressing emotions, but serve to communicate behavioral intentions or more generally social motives. According to this view, NVB communicates how people are likely to act, rather than how they are currently feeling. For the purpose of the present research, we will not go into this debate in detail (see Parkinson 2005 for a review) as both accounts make similar predictions regarding score estimations in sport competitions based on NVB: “Indeed, many emotions are precisely forms of social motive, and many social motives are emotional” (Parkinson 2005, p. 301).

Predicting Outcomes from Thin Slices of Expressive Behavior

A line of research providing evidence for the adaptive importance of nonverbal communication—which has been termed the *thin slices* approach—demonstrates that people are highly accurate at predicting various outcomes from short observations of NVB (Ambady et al. 2000; Ambady and Rosenthal 1992): including personality characteristics (Borkenau et al. 2004), conversation outcomes (Curhan and Pentland 2007), teaching effectiveness (Ambady and Gray 2002), leadership competence (Rule and Ambady 2008), and sales effectiveness (Ambady et al. 2006). In line with the reviewed evolutionary accounts of NVB, accurate positive versus negative judgments of NVB that are especially important to survival (e.g., dominance, fear, disgust) can be made within seconds, whereas longer exposure times are needed when making more complex judgments, e.g., of personality variables (Carney et al. 2007).

Hence, the ability to quickly and reliably detect certain attributes such as anger, fear, dominance, or submissiveness evolved because they are essential for survival in social animals by preparing the organism to act adaptively. Moreover, the ecological approach (Fridlund 1994, see also McArthur and Baron 1983) emphasizes that accurate social perception is not limited to its utility in detecting emotions, but also serves adaptive functions at the level of individual goal attainment (e.g., whom one should better avoid in a confrontational situation).

The Present Research

Of relevance to the present study, Booth et al. (1989) demonstrated that winning in a competition raises testosterone levels, which has been linked to the nonverbal display of dominance (Carney et al. 2010). In addition, losing in competition had the opposite effect of lowering testosterone which is associated with nonverbal displays of submissiveness. In combination with the findings from the thin slices of expressive behavior approach, it therefore seems feasible that leading and trailing in ongoing sport competitions would result in subtle nonverbal behavioral changes associated with dominance and submissiveness, which are reliably recognized by observers and competitors. This argumentation is anecdotally supported by the frequent referral of sport commentators to the NVB of athletes as a consequence of either leading very clearly or trailing by far. To our knowledge, currently no research exists on the nonverbal expression changes associated with leading or trailing during competition and, more importantly, whether people can accurately interpret these NVB as signs of leading or trailing.

Based on the reviewed evolutionary accounts of NVB (e.g., Shariff and Tracy 2011), we hypothesized that people—including adults with no knowledge of the respective sport (Experiment 1), children (Experiment 2), and athletes (Experiment 3)—will be able to judge who is leading or trailing in ongoing sport competition. In this endeavor, we transferred the thin slices paradigm to the field of sports and thereby further addressed a well-known difficulty of this approach: objectively defining accuracy of predictions based on the observation of thin slices of expressive behavior (cf. Ambady and Rosenthal 1992). Accuracy typically describes the correspondence between a judgment and a criterion (Kruglanski 1989) which has proven to be problematic in social psychology, as objective and externally valid criteria against which to evaluate predictions are difficult to find. In this respect, we consider the sports context as highly suitable to investigate the predictive power of certain nonverbal cues on performance outcomes, as the outcome of sports competitions is objectively defined by scores.

In three experiments we tested the hypothesis that score estimates of perceivers would correspond with the actual scores when viewing short video footage of sporting competition. In Experiment 1, we tested this hypothesis in the sports of basketball and table tennis amongst perceivers who had no experience in the respective sports. In Experiment 2, two groups of children (4–8 and 9–12 years) were tested with the same experimental procedure as in Experiment 1. Based on recent findings (Balas et al. 2012; Ross et al. 2012; Thomas et al. 2007), we hypothesized that children would be able to distinguish between leading and trailing athletes, but that their estimations would be less accurate than adults' estimations. In Experiment 3 we followed a recent call of Fiedler (2011) for replicating an effect with different stimuli, while further examining whether domain-specific knowledge in a respective sport moderates this effect. According to the reviewed evolutionary accounts of NVB both inexperienced perceivers and experienced athletes should be equally well-equipped to recognize specific NVB and interpret these as signs of leading or trailing.

Experiment 1: Who's Leading or Trailing

Method

Participants

Forty college students (20 male and 20 female; $M = 22.4$; $SD = 1.4$) took part in the study, who were not experienced in both respective sports and stated they had never played these sports in an organized context. Neither gender nor age significantly influenced the pattern of results. Written informed consent was obtained from every participant before commencing the experiment. The study was carried out in accordance with the Helsinki Declaration of 1975.

Stimuli

We selected video footage of televised basketball games from the NBA and the highest German league (Seasons 2010–2012) and table tennis matches from the World Cup, the European Cup, the Chinese Super League, and the highest German League. The following selection criteria were implemented to assure that the chosen stimuli were suitable for testing the hypothesis that leading and trailing in ongoing competition result in specific NVB that resemble evolved adaptations and are therefore recognized by observers: In order to ensure that the ratings were not influenced by score-induced changes in sport specific behavior such as strategy or tactics, we chose video stimuli that involved breaks during the game—including time-outs, free throws in basketball, and breaks between points in table tennis. Further, we carefully avoided showing any kind of obvious non-verbal signals associated with pride such as raising both fists above the head or obvious signals displaying shame such as hiding the face behind the hands (cf. Tracy and Matsumoto 2008), that have empirically been linked to the final outcome in sport and therefore would be too informative cues for estimating the score. After the selection process was complete, selected basketball videos had a mean duration of 3.9 s ($SD = 2.8$; $Mode = 1$) and selected table tennis videos had a mean duration of 3.5 s ($SD = 3$; $Mode = 1$).

Experimental Manipulation

The manipulation in the study involved the actual score of the game during the video. We chose five different categories of scores: (1) far behind, displaying a team or player trailing substantially, which was defined in basketball as at least fifteen points behind and in table tennis as at least five points behind. Moreover, in this category the team or player shown always lost the game in the end; (2) close behind, showing a team or player losing in a fairly close game situation which was defined in basketball as no more than five points behind and in table tennis no more than two points behind; (3) a draw in which the score was equal; (4) close lead, showing a team or player leading in a fairly close game situation which was defined in basketball as no more than five points ahead and in table tennis no more than two points ahead; (5) high lead, displaying a team or player leading substantially, which was defined in basketball as at least fifteen points ahead and in table tennis as at least five points ahead. Moreover, in this category the team or player shown always won the game in the end.

We aimed at finally having a battery of 20 videos in each experimental category for both respective sports (200 video clips in total). The selection of videos was done by student research assistants according to the following guidelines. Research assistants were instructed to review all videos of a convenience sample of 30 basketball games and 40 table tennis games one after the other. They were to select each video that fit the above mentioned criteria [breaks during game, no obvious nonverbal signals that have empirically been linked to victory and defeat (Tracy and Matsumoto 2008)], until each category of scores contained 20 videos¹ (see footnote for the hyperlink to the utilized stimulus material).

Measure

Perceivers rated the short video scenes on an 11-point digital semantic differential scale after every video. In order to give their ratings, perceivers moved a mouse cursor from the middle of the scale which represented a tied score towards either pole of the scale and logged in their rating by clicking the left mouse button. The software converted the ratings into a value (with 3 decimals) between 0 reflecting the left pole of the scale with the label “*far behind*” and 1 reflecting the right pole of the scale with the label “*high lead*.” The utilized scale was continuous, ranging from 0.000 to 1.000 and was visually presented as 11 points in order to assist participants in providing a clear indication of their ratings.

Procedure

Perceivers were instructed that they had to estimate who was leading or trailing based on the video footage presented to them by moving a mouse cursor to either the “*high lead*” or “*far-behind*” pole of the semantic differential scale. They were further instructed to answer as accurately as possible, while speed was not emphasized. Before commencing the experiment, perceivers filled out a questionnaire gathering demographic data. Every

¹ In order to maximize transparency in the conducted research we provide hyperlinks to the stimulus material utilized in the studies. Note that the software randomly selected and displayed the stimulus material from the video stream according to the described procedure and not as shown in the video streams: basketball: (<http://www.youtube.com/watch?edit=vd&v=UsviKNsOkUM>), table tennis: (<http://www.youtube.com/watch?v=2Y3YeYqTnSY>), handball: (<http://www.youtube.com/watch?v=w4zuaDe9dzs>).

perceiver was tested individually on a standard 17 inch notebook placed 60 cm away from the perceivers. E-prime 2.0 professional (Psychological Software Tools 2007) was used to present the stimuli and collect the judgments. All videos were presented silently to ensure that ratings were based on NVB and not, for example, crowd noise. For every perceiver the software randomly chose 12 videos from the categories far behind and high lead for both basketball and table tennis. For the other three categories close behind, draw, and close lead only four videos for each sport were randomly chosen so that twelve videos were also presented for the combined category “close score.” Hence, every perceiver viewed 72 videos out of the 200 video clip battery in random order. In other words, different perceivers were randomly assigned to different sets of videos. This approach helps to ensure that results do not depend on specific combinations of stimuli. After every video clip perceivers had to give their rating by clicking the left mouse button on the score estimation scale described above. After completing the testing procedure, participants were informed about the purpose of the experiment.

Results

The descriptive results of Experiment 1 are displayed in Fig. 1. A 2 (basketball vs. table tennis) $\times$ 5 (far behind, close behind, draw, close lead, and high lead) ANOVA revealed a significant main effect of sport on overall score estimates [$F(1, 39) = 22.734, p < .001, \eta^2_p = .368$] indicating overall higher ratings—closer towards the “high lead” pole of the scale—in the sport of table tennis. Most importantly, there was a significant main effect of actual score on the score estimates [$F(3.272, 127.619) = 35.257, p < .001, \eta^2_p = .475$] indicating that perceivers were very accurate at estimating whether both basketball and table tennis players were leading or trailing. In addition, there was a significant interaction between sport and actual score [$F(2.814, 109.727) = 8.369, p < .001, \eta^2_p = .177$].

Follow-up polynomial linear contrasts across both sports revealed a strong linear relationship between the score estimates and the score categories [$F(1, 39) = 259.40, p < .001, \eta^2_p = .869$] demonstrating that the score estimates corresponded in a linear manner with the scores during the game. The significant interaction in the overall ANOVA is explained by the larger effect size in the linear contrast analysis for the basketball stimuli [$F(1, 39) = 195.881, p < .001, \eta^2_p = .834$] compared to the table tennis stimuli [$F(1, 39) = 77.180, p < .001, \eta^2_p = .664$].

Discussion

Across two different sports, inexperienced perceivers, with no domain specific knowledge regarding basketball or table tennis, as assessed via self-ratings, were able to estimate the tendency of who was leading and who was trailing based on thin slices of NVB. In accordance with Clark-Carter (1997) who suggested that η^2_p values above 0.138 may be classified as large effects, the main effect of actual score on the score estimates $\eta^2_p = .475$ observed in Experiment 1 represents a large statistical effect. This finding is in line with our hypothesis derived from evolutionary accounts of NVB (Fridlund 1994; Shariff and Tracy 2011) and the thin slices literature (Ambady et al. 2000; Ambady and Rosenthal 1992). Hence, communicating internal states nonverbally—in this case nonverbal signals associated with leading and trailing in sporting competitions—seems to be readily recognized and understood among adults (Burgoon 1996).

Although the findings of Experiment 1 might be interpreted as being in line with evolutionary accounts of NVB, they do not necessarily suggest that reliable decoding of

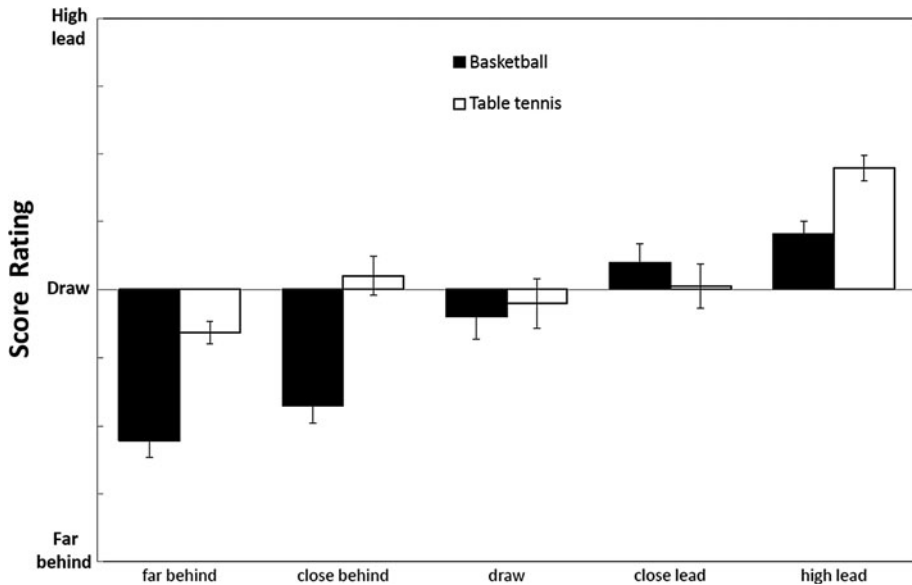


Fig. 1 Mean score (0 = *far behind*/high lead) estimates as a function of score category and sport. Error bars represent standard errors

nonverbal cues represents a fixed capacity that humans are born with. Recent evidence (Batty and Taylor 2006; Durand et al. 2007; Gao and Maurer 2010; Ross et al. 2012; Tonks et al. 2007) suggests that decoding of nonverbal signals has both a nature and a nurture component. Hence, interpreting nonverbal cues as indicative of leading or trailing might further depend on learning, experience, and brain maturation processes (Balas et al. 2012; Thomas et al. 2007). Therefore, we replicated Experiment 1 with a group of 4–8 year old and 9–12 year old children following a similar between-group approach as Balas et al. (2012).

Experiment 2: Leader–Trailer Judgments in Children

The disentanglement of the relative contributions of nature and nurture in the nonverbal encoding and decoding of internal states is complicated by the fact that children are exposed to complex emotional experiences from their birth. However, there is increasing evidence suggesting that the nonverbal recognition of emotions continues to develop until adulthood (Kolb et al. 1992; Tonks et al. 2007). These findings are interpreted as developmental adaptations to the increasing demands of the social environment as humans grow older. Therefore, adults show a more sensitive fine-tuning regarding both differentiation and reliably judging the intensity of different emotions (Ross et al. 2012). The authors assume that ongoing brain maturation contributes to this finding as brain regions associated with social and emotional perception undergo developmental structural changes at least until the end of the second decade of life. In addition, Balas et al. (2012) provided first evidence that the reliable judgments from thin slices of NVB only develop slowly.

Therefore, we hypothesized that children would be able to distinguish between leading and trailing athletes, but that their estimations would be less accurate than adults' estimations as obtained in Experiment 1.

Method

Participants

Forty-four children (23 male and 21 female) took part in the study. In line with Balas et al. (2012) we divided the participants in different age groups. Twenty-two children made up the 4–8 year old group (11 male and 11 female; $M = 6.7$ years; $SD = 1.4$) and another twenty-two the 9–12 year old group (12 male and 10 female; $M = 11.1$ years; $SD = 0.8$). Gender did not significantly influence the pattern of results. Written informed consent was obtained from the parents of every participant before commencing the experiment. The study was carried out in accordance with the Helsinki Declaration of 1975.

We utilized exactly the same stimuli and experimental procedure as in Experiment 1.

Results

The descriptive results of Experiment 2 are displayed in Fig. 2. A 2 (4–8 year vs. 9–12 year) $\times$ 2 (basketball vs. table tennis) $\times$ 5 (far behind, close behind, draw, close lead, and high lead) ANOVA again revealed a significant main effect of actual score on the score estimates [$F(4, 168) = 9.800, p < .001, \eta^2_p = .189$] indicating that irrespective of the age group and stimulus material children were able to estimate who was leading or trailing based on NVB. No other main effects or interactions reached significance (all $p < .32$). Interestingly, the interaction between age group and actual score [$F(4, 168) = .421, p = .793, \eta^2_p = .01$] was not significant and did not show any trend that the older children were better able to judge who was leading or trailing (cf. Fig. 2). In Experiment 2, the sport from which the stimulus material was selected did not have any influence on the ratings. The young participants were equally likely to correctly decode the NVBs in both sports as signs of leading and trailing.

Polynomial linear contrasts across both sports again revealed a strong linear relationship between the score estimates and the score categories [$F(1, 42) = 58.40, p < .001, \eta^2_p = .582$] for both the younger and older children (cf. Fig. 2). The linear contrast analysis demonstrated similar effects for both the younger [$F(1, 21) = 36.850, p < .001, \eta^2_p = .637$] and older children [$F(1, 21) = 25.153, p < .001, \eta^2_p = .545$].

Comparison Between Adults and Children

In order to compare the adults' results from Experiment 1 with the children's results from Experiment 2 (see Fig. 3 for an overview), we calculated confidence intervals for the effect sizes of the polynomial linear contrasts that depict the linear relationship between the score estimates and the score categories (Cumming 2012). In order to do so, we first calculated the polynomial linear contrasts collapsed across both sports for adults ($\eta^2_p = .870$) and for children ($\eta^2_p = .583$). We transformed partial eta-square into the effect size r using the calculations provided by Cohen (1988) and then calculated the 95 % confidence intervals for both effect sizes (Cumming 2012). Confidence intervals of the adults' effect size

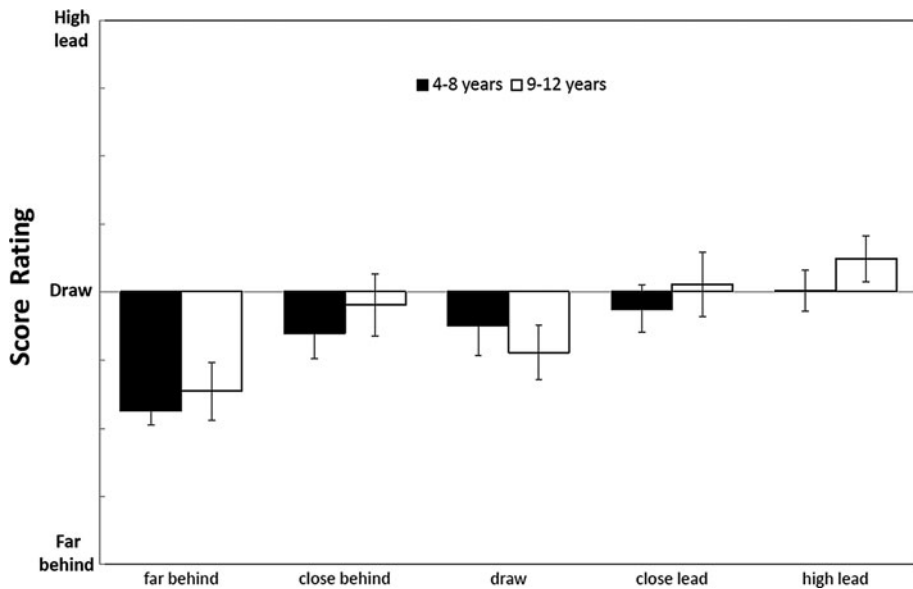


Fig. 2 Mean score (0 = *far behind* to *high lead*) estimates as a function of score category and child age group. Error bars represent standard errors

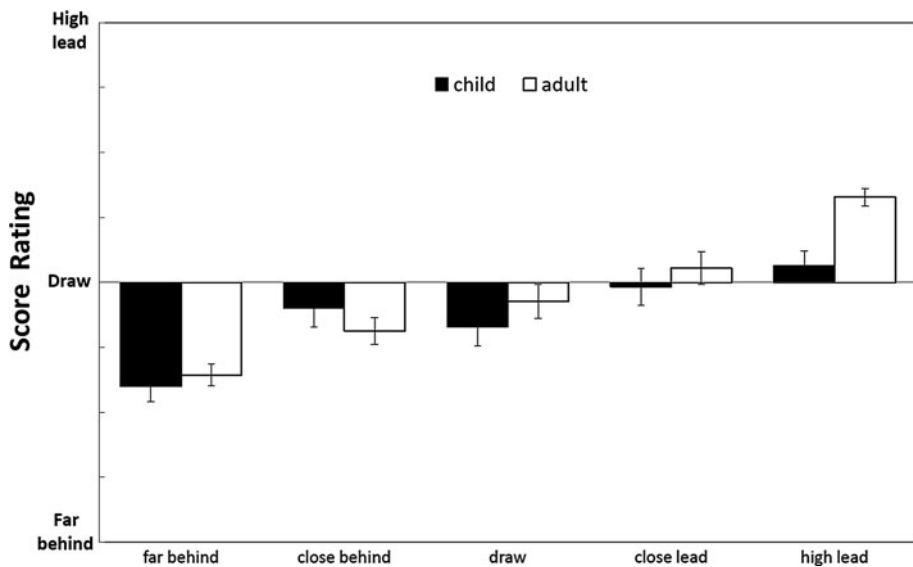


Fig. 3 Mean score (0 = *far behind* to *high lead*) estimates as a function of score category and age group. Error bars represent standard errors

[$r = .93$, 95 % CI (.871, .963)] and the children's effect size [$r = .76$, 95 % CI (.598, .862)] do not overlap. This finding suggests that adults' score estimations corresponded to a higher degree with the actual score than children's estimations (cf. Fig. 3).

Discussion

The results from Experiment 2 showed that children between 4 and 12 years of age were able to reliably distinguish between athletes who were trailing by far from athletes who were leading in the sports of basketball and table tennis. Again, the main effect of actual score on the score estimates $\eta^2_p = .189$ observed in Experiment 2 represents a large statistical effect and is in line with our hypothesis derived from evolutionary accounts of nonverbal expressions (e.g., Shariff and Tracy 2011) and the thin slices literature (Ambady et al. 2000; Ambady and Rosenthal 1992). Older children (9–12 years) were no more accurate at estimating who was leading or trailing based on NVB than young children (4–8 years).

However, and in line with previous findings (Balas et al. 2012; Ross et al. 2012; Thomas et al. 2007), adults' score ratings corresponded to a higher degree to the actual score of the experimental stimuli, which was evident in a larger effect size in the linear contrast analysis for adults. This finding suggests that the fine-tuning (Thomas et al. 2007) of distinguishing between subtle nonverbal cues continues to develop until adulthood. Previous theorizing has suggested that this finding might be explained by the increasing demands of the social environment (Batty and Taylor 2006; Durand et al. 2007; Gao and Maurer 2010; Tonks et al. 2007) and by brain maturation processes (Batty and Taylor 2006; Thomas et al. 2007, 2001). The fact that we only found a difference between adults and children and not between younger and older children might be interpreted as in line with the suggestion that adolescence is a highly important developmental phase in the decoding of NVB (Ross et al. 2012).

As the results of Experiments 1 and 2 suggest, that experience seemed to influence the effect of estimating who was leading and trailing based on NVB, we attempted to further investigate whether domain specific sport knowledge would influence this effect in Experiment 3. As we only controlled for experience in playing the respective sports in Experiment 1 and not TV exposure, which might mean that the participants were not entirely ignorant of table tennis or basketball, we controlled for sport TV exposure additionally in Experiment 3. Furthermore, we followed a recent call by Fiedler (2011), who pointed out the necessity of replicating effects found with one set of stimuli with different stimuli to ensure that the phenomenon of interest does not only apply to a highly specific set of stimulus material, but applies generally for the phenomenon of interest.

Experiment 3: Experience based Differences in Estimating Who's Leading or Trailing

Method

Participants

Altogether 40 participants took part in Experiment 3. Twenty college students (10 male and 10 female; $M = 23.6$; $SD = 2.28$) reported having no experience in playing handball and not following handball on television ($M_{\text{hours of handball on TV/week}} = 0.09$; $SD = 0.19$). The other twenty participants were competitive handball players who were competing at an amateur to semi-professional level in Germany (10 male and 10 female; $M_{\text{age}} = 25.3$; $SD = 4.38$; $M_{\text{years of handball experience}} = 17.5$; $SD = 5.44$; $M_{\text{hours of handball on TV/week}} = 1.5$; $SD = 1.65$). Two independent *t* tests verified that both groups differed significantly in both average years of handball experience ($t(38) = 13.94$, $p < .001$, two-tailed) and average

TV exposure of handball per week ($t(38) = 3.778$, $p = .001$, two-tailed). Hence, the samples chosen were suitable for investigating whether domain-specific sports knowledge influences the score estimates. Otherwise, neither gender nor age significantly influenced the pattern of results. Written informed consent was obtained from every participant before commencing the experiment. The study was carried out in accordance with the Helsinki Declaration of 1975.

Stimuli

We selected video footage of televised handball games according to similar criteria as in Experiment 1 from a convenience sample of 20 games of the European Championships of men in 2012 and the World Championships of women in 2011. Again, we only chose scenes that involved breaks during the game—including time-outs or situations in which the ball was out of bounds. In addition, the scenes did not show any of the obvious nonverbal signals that have been empirically linked to victory and defeat in sport (cf. Tracy and Matsumoto 2008). Although it seems unlikely, as participants in Experiment 1 were not familiar with the respective sports, a possible alternative explanation for the results is that participants used actual playing level or reputation of teams or athletes (or some other variable that is correlated with playing level such as reputation) as cues for their estimates. In order to rule out this possibility in Experiment 3, we made sure that the actual playing level of the team shown was not confounded with the score category of the video. Therefore, we checked that the placing of the teams in the respective tournaments was equally distributed in all experimental categories. After selection, videos had a mean duration of 4 s ($SD = 2$; $Mode_1 = 2$; $Mode_2 = 3$). An additional variation to Experiment 1 stemmed from the specific rules of team-handball which result in fewer regular breaks during the game than in basketball and table tennis. Therefore, we decided to not only depict athletes' NVB but also the NVB of coaches during competition in order to obtain an equal amount of videos from our sample of games as in Experiment 1.

Experimental Manipulation

We chose the same five categories as in Experiment 1: (1) far behind, showing a team that is at least seven goals behind and always lost the game in the end; (2) close behind, showing a team that is exactly two goals behind; (3) a draw in which the score was equal; (4) close lead, showing a team that is exactly two goals ahead; (5) high lead, a team at least seven goals ahead and always winning the game in the end. Videos were again selected by student research assistants. Again, we aimed at finally having 20 videos in each score category (see Footnote for the hyperlink to the utilized stimulus material). Selection guidelines were the same as in Experiment 1, except that the research assistants could also include videos depicting the coach.

Procedure and Measure

The procedure and measure used in Experiment 3 was similar to Experiment 1 and only the changes are described. In Experiment 3, the stimulus material was taken exclusively from handball. The software randomly chose twelve videos from the categories far behind and high lead. For the other three categories close behind, draw, and close lead only four videos were randomly chosen so that twelve videos were also presented for the combined category

“close score.” Hence, every perceiver viewed 36 videos out of the 100 video clip battery in random order.

Results

The descriptive results of Experiment 3 are depicted in Fig. 4. The 2 (handball player vs. naïve) $\times$ 5 (far behind, close behind, draw, close lead, and high lead) ANOVA most importantly revealed a significant main effect of actual score on the score estimates [$F(4, 152) = 27.489, p < .001, \eta^2_p = .420$], indicating that participants were accurate at estimating whether the depicted team was leading or trailing. There was neither a significant main effect for handball experience [$F(1, 38) = 3.290, p = .078, \eta^2_p = .080$] on score ratings, nor did handball experience interact with the actual score categories of the videos [$F(4, 152) = 0.524, p = .718, \eta^2_p = .014$]. This suggests that both participant groups were equally able to infer who was leading or who was trailing from the nonverbal displays of the athletes.

Polynomial linear contrasts showed a strong linear relationship between the score estimates and the score categories [$F(1, 38) = 117.181, p < .001, \eta^2_p = .755$] for both the handball players and non-expert participants (cf. Fig. 4). The linear contrast analysis even indicated slightly larger linear effects for the unexperienced [$F(1, 19) = 93.108, p < .001, \eta^2_p = .831$] than for the handball players [$F(1, 19) = 47.181, p < .001, \eta^2_p = .713$].

Discussion

We were able to replicate the pattern of results from Experiment 1 in the sport of handball, demonstrating that people seem to be able to differentiate between leading and trailing athletes based on NVB. The statistical effect observed ($\eta^2_p = .420$) in Experiment 3 was

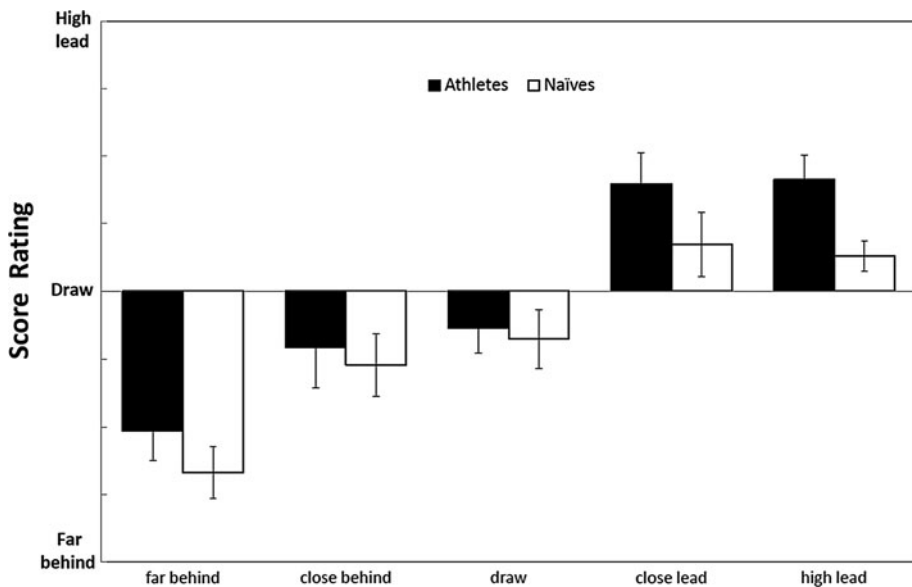


Fig. 4 Mean score (0 = far behind, 100 = high lead) estimates in basketball as a function of score category and sport experience. Error bars represent standard errors

similar to Experiment 1 ($\eta^2_p = .475$) and confirms that nonverbal expressions associated with leading and trailing have a large effect (Clark-Carter 1997) on estimating the score. In addition, domain specific knowledge did not influence this finding, as both proficient handball players and unexperienced students rated the experimental categories similarly. This finding is in line with the assumption of evolutionary accounts of NVB (Darwin 1872; Shariff and Tracy 2011) that humans have evolved the ability to perceive emotions (Ekman 1992) and other internal states (Fridlund 1994) communicated by another person nonverbally.

Of further interest, the main effect of actual score on score estimates was significant when only analyzing the clips (45 clips) that primarily depicted the coaches' NVB ($p < .001$, $\eta^2_p = .220$) and the clips (55 clips) that primarily depicted the players' NVB ($p < .001$, $\eta^2_p = .265$). This suggests that both coaches and players alter their NVB as a function of leading and trailing in ongoing competition that can be used as cues for the actual score. However, this post hoc analysis is problematic as we did not treat this as an experimental factor in the planning of our research method. Therefore, the ratios of coach versus player stimuli were not evenly distributed across the experimental categories. In addition, the random selection of the E-prime software was not programmed to display the same number of coach and player videos. Hence, although interesting, future research will have to confirm this initial finding.

General Discussion

The aim of the present study was to investigate whether people are able to detect who is leading or trailing in an ongoing sport competition based solely on nonverbal cues. The present research was the first to show that leading and trailing in sports are associated with nonverbal cues that are interpreted as signs of winning and losing. Thin slices of behavior (very short recordings of athletes' and coaches' behavior) were sufficient to permit quite precise estimates of the actual score in sport competitions. We were able to obtain this finding with stimuli from both individual and team sports, with stimuli depicting athletes and stimuli depicting coaches, amongst adults and children, and amongst observers with and without experience in the respective sport. Whilst domain-specific knowledge did not influence the effect, age did. Together with previous findings (Balas et al. 2012; Batty and Taylor 2006; Durand et al. 2007; Gao and Maurer 2010; Ross et al. 2012; Thomas et al. 2007; Thomas et al. 2001; Tonks et al. 2007), the fact that we only found a difference between adults and children, and not between younger and older children, suggests that adolescence might be a highly important developmental phase in the decoding of nonverbal behavior.

The findings presented were not only in line with evolutionary accounts of NVB (Fridlund 1994; Shariff and Tracy 2011) and the thin slices of behavior hypothesis (Ambady et al. 2000; Ambady and Rosenthal 1992), but they add significantly to the literature on nonverbal behavior in sports. So far, research on nonverbal behavior in sports has either focused on post-performance emotional expressions (Matsumoto and Willingham 2006) or it has focused on pre-performance NVB (Furley and Dicks 2012; Furley et al. 2012a, 2012b; Greenlees et al. 2005a, 2005b). In contrast, our research suggests that NVB occurring during different stages of the game can be interpreted as cues as to who is currently leading and who is trailing.

One strength of the present research was that we were able to obtain the predicted findings in three different studies (altogether 124 participants) with stimuli from three

different sports. Among these were both individual (table tennis) and team sports (basketball, handball). Furthermore, not all observers were shown the same stimuli, but different participants were randomly assigned to different subsets of all stimuli. These approaches are likely to reduce the probability that effects were dependent on one particular set of highly selected stimuli, as recently called for in psychological research (Fiedler 2011). It is also in line with the calls for replication that are becoming more frequent in the psychological literature (Pashler and Wagenmakers 2012; Yong 2012). Therefore, it is our further hope that other research groups will be stimulated to replicate and extend these first findings.

The finding that laypeople and young children were able to distinguish between leading and trailing athletes based on NVB provides support for the evolutionary nature of NVB and its interpretation (Fridlund 1994; Shariff and Tracy 2011). Although the findings can be interpreted as being in line with evolutionary accounts of NVB, the comparison between children and adults suggests that the decoding of nonverbal cues has both a nature and a nurture component. On the one hand, a line of previous research suggests that the display of NVB as a consequence of success and failure has a strong nature component. In this respect, Tracy and Matsumoto (2008) found that sighted, blind, and congenitally blind athletes in Olympic competitions show similar NVB, although the latter athletes could not have been socialized to display these NVB. On the other hand, the present findings, in combination with the reviewed developmental studies, suggest that the fine-tuning of interpreting nonverbal signals also has a nurture component. Nevertheless, future research is needed on the relative contributions of nature and nurture in both the encoding and decoding of nonverbal signals.

A further strength of the present approach is its high external validity. As the stimuli are not artificially created—which is typical for the majority of research on NVB in sports (Furley et al. 2012a, b; Greenlees et al. 2005a, b)—but recordings of real sports competitions, the results are likely to transfer to the field. Hence, people are likely to accurately infer who is currently leading or trailing from athletes' NVB. Using true life stimuli provides the additional advantage that the *accuracy* of participants' estimates can be assessed based on an objective criterion, the score. Our findings thereby add to the thin slices literature by remedying a common problem of this paradigm, namely that the criterion for predictions cannot be objectively assessed (Ambady et al. 2000; Ambady and Rosenthal 1992).

The main limitation of our research so far concerns potential selection bias with regard to the collection of the stimulus material, which is a common problem in psychological literature (Fiedler 2011). Although we paid careful attention to this point when selecting the stimulus material, we cannot entirely rule out the possibility that scores are not only correlated with NVB, but confounded or influenced with some third variable that is displayed in the video clips. If this were true, participants would not only estimate scores based on NVB, but on some other variable. We consider this possibility to be rather small, as we found effects for three different sets of video clips.

In addition, some athletes were present in several score categories, and it seems feasible that differentiating between leading and trailing would be easier for the same athlete than for different athletes. Further, potential problems in the stimulus material might be that some athletes could be more expressive than others, or that some video clips contained more information than others. Hence, we cannot entirely rule out that certain stimuli or constellations of stimuli mainly account for the pattern of results. We tried to partially address this issue by our random selection procedure from our stimuli battery. Nevertheless, these issues highlight important avenues for future research. In order to maximize

transparency in the research conducted, we provide hyperlinks to the stimulus material utilized in the studies (see footnote). Hence, readers can judge for themselves the likelihood of potential alternative explanations resulting from stimulus selection. In this respect, we follow a recent call by Pashler and Wagenmakers (2012) on how to reduce the “crisis of confidence in psychological science” (p. 1).

While answering important questions, our research also poses new ones, both on a theoretical and an applied level. These questions refer to the functionality of the phenomenon we described in competitive sports. According to evolutionary accounts of nonverbal expressions, submissive nonverbal expressions originally served the function of avoiding further attacks in a fight (Shariff and Tracy 2011). This seems highly functional during a fight between primates (or other animals), as the winning competitor is likely to stop hurting the opponent any further when perceiving submissive NVB. However, the story might be quite different in competitive sports: Here, showing “losing-NVB” might be highly dysfunctional, as the winning opponent is not likely to hold off. When perceiving “losing-NVB”, the winning opponent might even increase pressure. This reasoning suggests that certain evolved behaviors may be dysfunctional in certain domains of modern life. What makes sense for a primate losing a fight may lead to exacerbating the downward spiral for athletes on the losing side.

Another interesting question concerns possible interactions between leading and trailing athletes. How do leading athletes react to the expression of submission by their losing opponent? This question directly leads to applied aspects of the present research. So far, we have shown that leading or trailing is correlated with the expression of certain NVB that other people can use in order to correctly infer who’s leading or trailing. However, we do not know how the NVB themselves may in turn influence further gameplay and the score. Athletes might recognize their opponents’ NVB and react accordingly—either with increased motivation (when leading) or apathy and decreased motivation (when trailing), for example. In line with this suggestion, previous research (Furley et al. 2012a, b) shows that athletes are more confident in competition when observing submissive or anxious NVB amongst their opponents prior to performance. As the perception of NVB seems to influence the self-efficacy of athletes (Furley et al. 2012a, b; Furley and Dicks 2012) and even their behavior (Furley et al. 2012a, b), the present findings may have important practical implications for sport teams and individual athletes that might for example help them avert the downward spiral of trailing in a competition.

In conclusion, the present research highlights that people are well-equipped to pick up subtle changes in NVB associated with leading and trailing in sport competitions. This seems to hold for individual and team sports, for experts and laypersons, and for children and adults.

Acknowledgments Special thanks go to Manuel Thiel, Christoph Haberl, Christoph Sobotta, and Wolfgang Walther for helping with the data collection and programming in this study. Further we would like to thank William D. Furley for copy editing the manuscript.

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